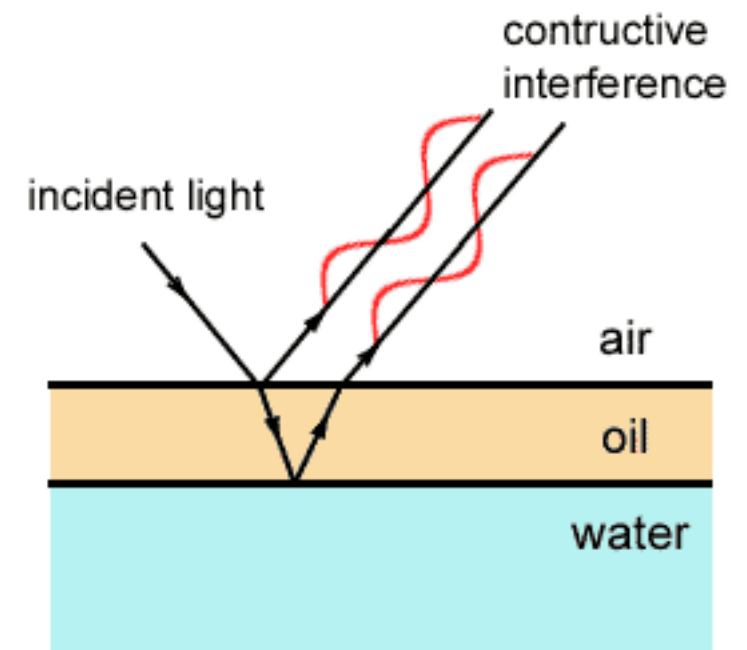


Oil spill in front of the reactor building..



Light and neutrons

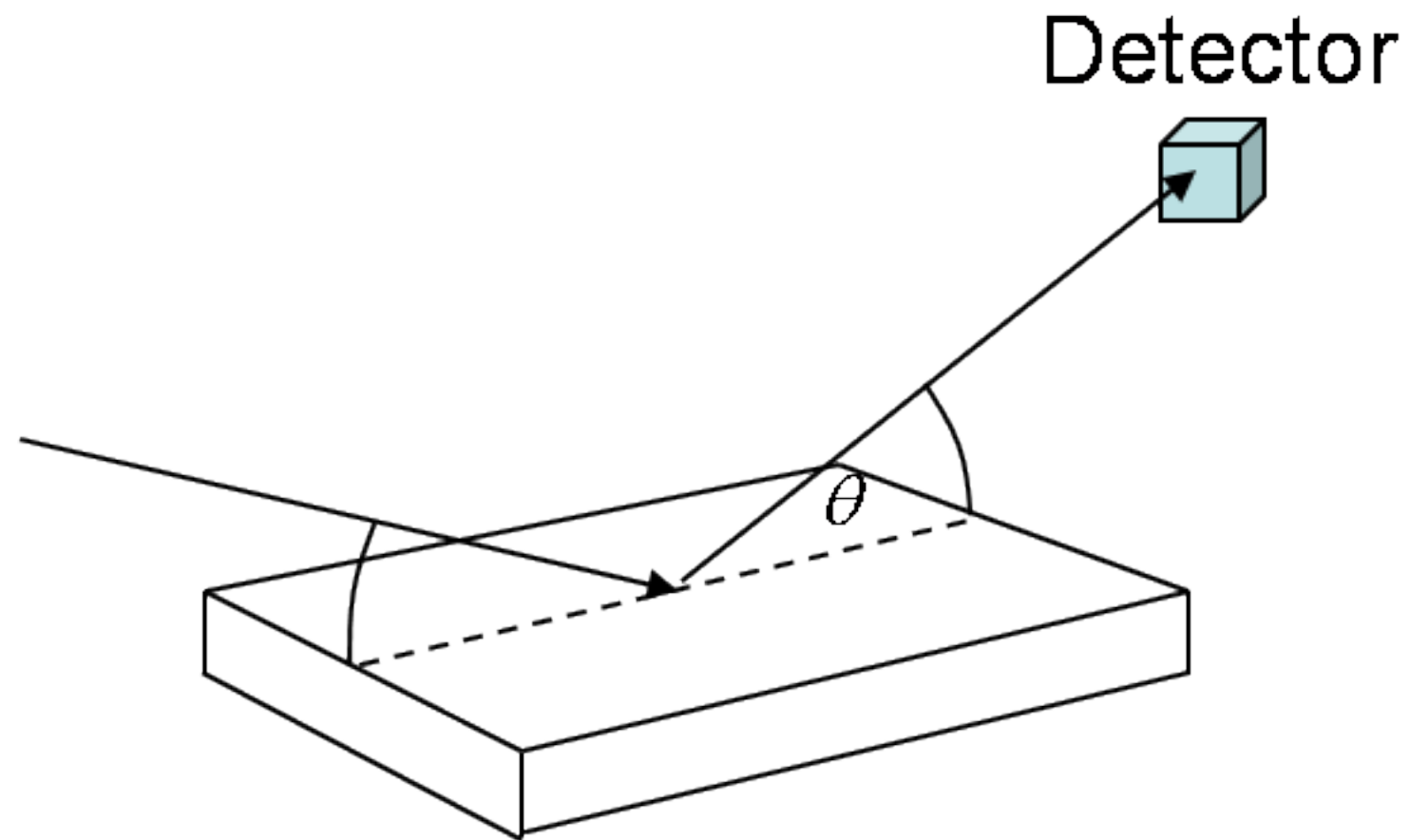
- Neutrons reflect from mirrors in a similar way as light.
- Manifestation of neutron's wave-particle nature!
- Light “sees” electrons while neutrons “see” nuclei

The underlying reason for these commonalities

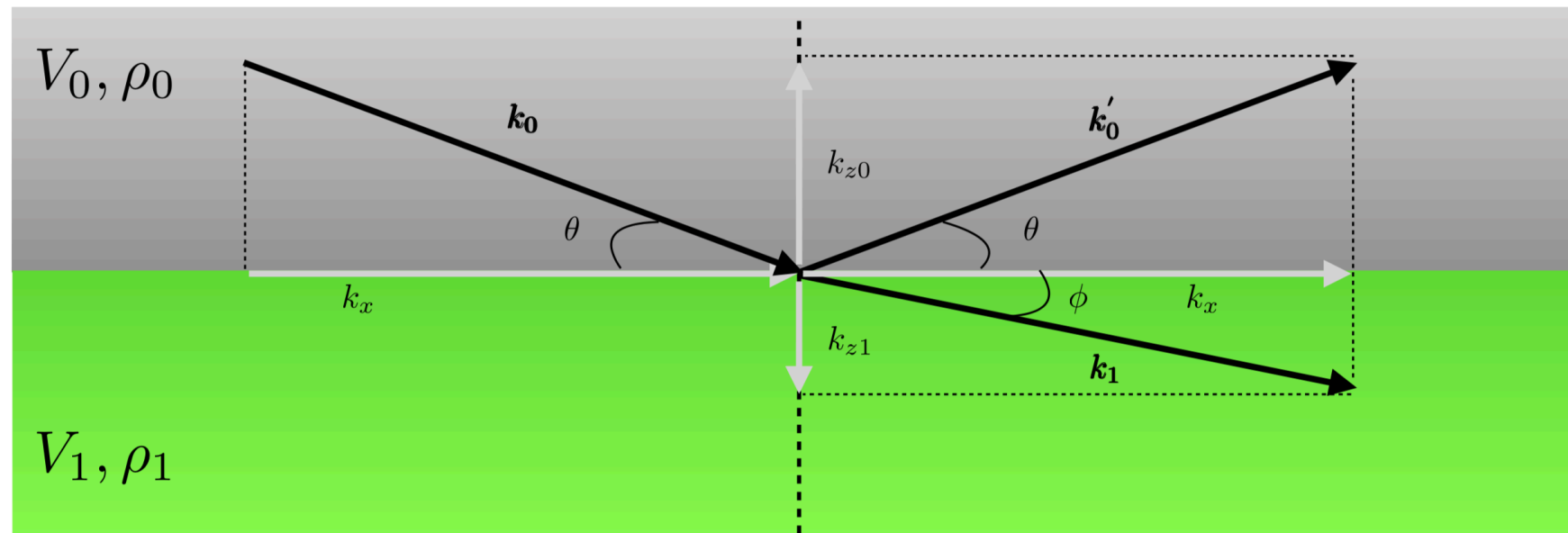
$$(\vec{\nabla}^2 + k^2)A = 0$$

both types of radiation their propagation obey a second-order partial differential equation of the same form (Helmholtz equation)

Neutron reflection at a planar surface



Neutron reflection at a planar surface



$$V = \frac{2\pi\hbar^2}{m}\rho$$

$$\rho = \sum_j N_j b_j$$

elastic specular reflection $|\vec{k}_0| = |\vec{k}'_0| = k = 2\pi/\lambda$

$$E = \hbar^2 k^2 / 2m$$

wave-vector transfer: $Q = |\vec{Q}| = |\vec{k}'_0 - \vec{k}_0| = 2k \sin \theta = 2k_z$

Starting point for calculations...

time independent Schrödinger equation for a particle in a constant potential

$$\frac{\hbar^2}{2m} \nabla^2 \psi(\vec{r}) + (E - V) \psi(\vec{r}) = 0$$

the interface is homogeneous in the xy-plane and reflection or refraction may take place only in the z-direction, thus problem reduced to one dimension

$$\frac{\hbar^2}{2m} \frac{d^2}{dz^2} \psi(z) + (E_z - V) \psi(z) = 0$$

$$E_{z0} = \frac{\hbar^2 k_0^2 \sin^2 \theta}{2m}$$

Neutron energy vs potential...

at small angles $V - E_z > 0$

large angles $V - E_z < 0$

for a critical angle we have $V - E_z = 0$

using $V = \frac{2\pi\hbar^2}{m}\rho$ and $E_{z0} = \frac{\hbar^2 k_0^2 \sin^2 \theta}{2m}$ we get

$$\cos \theta_c = \sqrt{1 - \frac{\lambda^2}{\pi}\rho} \quad \text{or} \quad \theta_c \approx \lambda \sqrt{\frac{\rho}{\pi}}$$

We solve the wave equation for the two cases...

for $E_z < V_1$ ($\theta < \theta_c$)

$$\frac{\hbar^2}{2m} \frac{d^2}{dz^2} \psi_0(z) + E_{z0} \psi_0(z) = 0 \quad \text{upper medium}$$

$$-\frac{\hbar^2}{2m} \frac{d^2}{dz^2} \psi_1(z) + \kappa^2 \psi_1(z) = 0 \quad \text{lower medium}$$

$$\text{with } \kappa^2 = \frac{2m}{\hbar^2} (V_1 - E_{z0})$$

solutions:

$$\psi_0(z) = e^{ik_{z0}z} + r e^{-k_{z0}z}$$
$$\psi_1(z) = t e^{-\kappa z}$$

Boundary conditions...

$$\left. \begin{array}{l} \text{wave function continuity: } \psi_0(0) = \psi_1(0) \\ \text{momentum conservation: } \frac{d\psi_0(0)}{dz} = \frac{d\psi_1(0)}{dz} \end{array} \right\} \begin{array}{l} 1 + r = t \\ ik_{z0}(1 - r) = -\kappa t \end{array}$$

$$\text{so } r = \frac{k_{z0} - i\kappa}{k_{z0} + i\kappa} \quad \text{and} \quad R = |r|^2 = \frac{|k_{z0} - i\kappa|^2}{|k_{z0} + i\kappa|^2} = 1$$

total reflection!

$$\text{also note: } |\psi_1(z)|^2 = |t|^2 e^{-2\kappa z} \quad \text{evanescent wave}$$

Second case...

for $E_z > V_1 (\theta > \theta_c)$

$$\frac{\hbar^2}{2m} \frac{d^2}{dz^2} \psi_{z0}(z) + \left(\frac{\hbar^2 k_{z0}^2}{2m} \right) \psi_{z0}(z) = 0 \quad \text{upper medium}$$

$$\frac{\hbar^2}{2m} \frac{d^2}{dz^2} \psi_{z1}(z) + \left(\frac{\hbar^2 k_{z0}^2}{2m} - \frac{2\pi\hbar^2}{m} \rho \right) \psi_{z1}(z) = 0 \quad \text{lower medium}$$

$$\text{or} \quad \left[\frac{d^2}{dz^2} + (k_{z0}^2 - 4\pi\rho) \right] \psi_1(z) = \left[\frac{d^2}{dz^2} + k_{z1}^2 \right] \psi_1(z) = 0$$

$$\text{with} \quad k_{z1}^2 = k_{z0}^2 - 4\pi\rho = k_{z0}^2 [1 - (\lambda^2/\pi)\rho]$$

solutions of wave equation...

$$\psi_0(z) = e^{ik_{z0}z} + re^{-ik_{z0}z}$$

$$\psi_1(z) = te^{ik_{z1}z}$$

plane waves moving in and out of the interface...

and by applying the same boundary conditions as previously we get

$$\left. \begin{array}{l} 1 + r = t \\ k_{z0}(1 - r) = k_{z1}t \end{array} \right\} \rightarrow \left. \begin{array}{l} r = \frac{k_{z0} - k_{z1}}{k_{z0} + k_{z1}} \\ t = \frac{2k_{z0}}{k_{z0} + k_{z1}} \end{array} \right\} \rightarrow \begin{array}{l} R = |r|^2 = \left| \frac{k_{z0} - k_{z1}}{k_{z0} + k_{z1}} \right|^2 \\ T = |t|^2 = \left| \frac{2k_{z0}}{k_{z0} + k_{z1}} \right|^2 \end{array}$$

we can work the last result a bit more...

$$k_{z0,c} = \frac{2\pi}{\lambda} \sin \theta_c$$

$$k_{z0,c}^2 = \frac{4\pi^2}{\lambda^2} \sin^2 \theta_c$$

$$k_{z0,c}^2 = \frac{4\pi^2}{\lambda^2} (1 - \cos^2 \theta_c)$$

remembering that $\cos \theta_c = \sqrt{1 - \frac{\lambda^2}{\pi} \rho}$

we get $k_{z0,c}^2 = 4\pi\rho$ and together with $k_{z1}^2 = k_{z0}^2 - 4\pi\rho$

we obtain $k_{z1}^2 = k_{z0}^2 - k_{z0,c}^2$

$$R = |r|^2 = \left| \frac{k_{z0} - k_{z1}}{k_{z0} + k_{z1}} \right|^2 \left. \vphantom{\frac{k_{z0} - k_{z1}}{k_{z0} + k_{z1}}} \right\} \rightarrow R = \left| \frac{k_{z0} - \sqrt{k_{z0}^2 - k_{z0,c}^2}}{k_{z0} + \sqrt{k_{z0}^2 - k_{z0,c}^2}} \right|^2$$

and finally we get...

$$\text{since } Q = 2k_{z0} \rightarrow R = \left| \frac{1 - \sqrt{1 - (Q_c/Q)^2}}{1 + \sqrt{1 - (Q_c/Q)^2}} \right|^2 \quad \text{exact expression}$$

$$\text{using the Taylor series } \sqrt{1 - (Q_c/Q)^2} = 1 - \frac{1}{2} \left(\frac{Q_c}{Q} \right)^2 + \dots$$

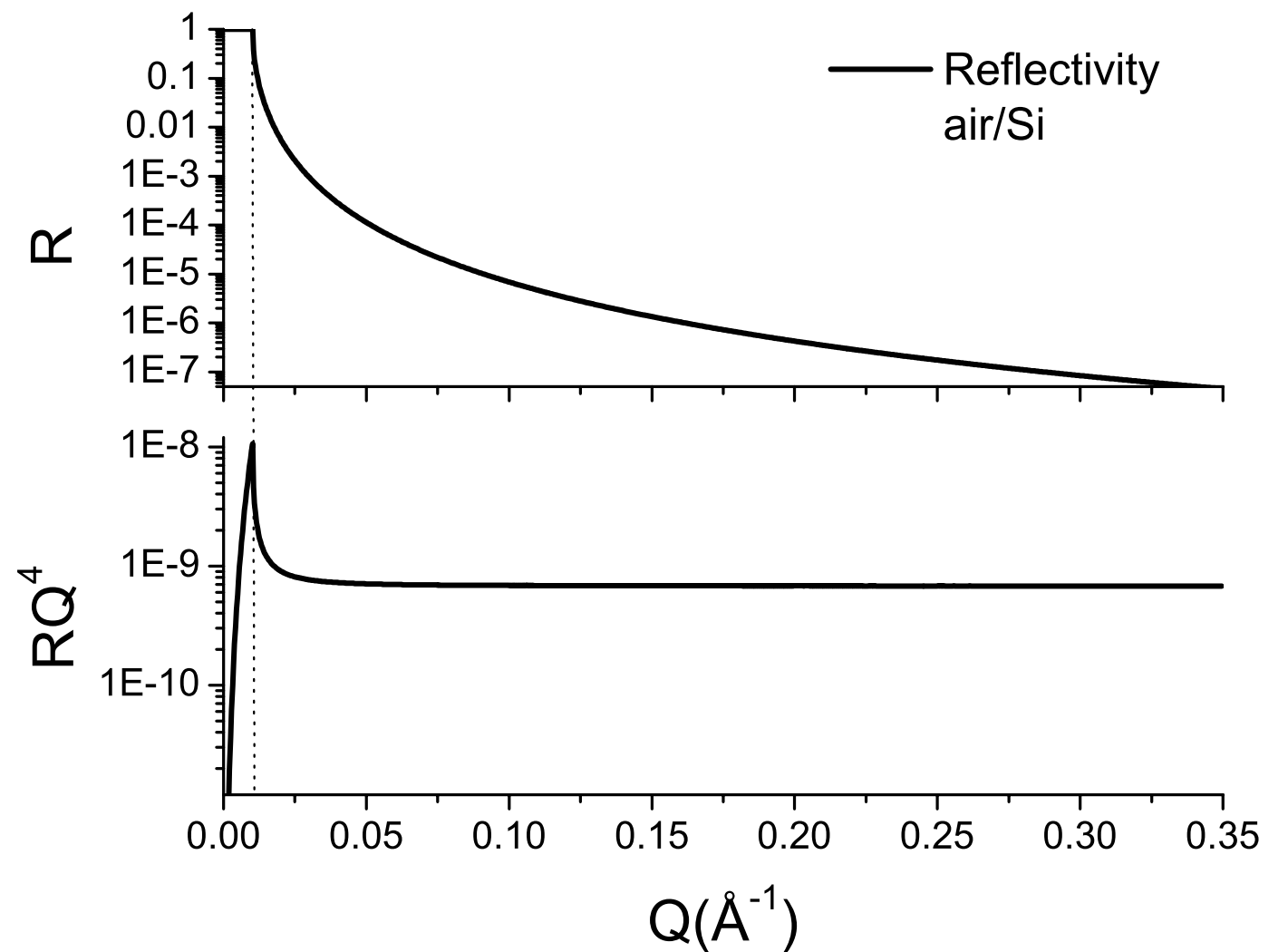
$$\text{for } Q \gg Q_c \quad R \approx \frac{1}{16} \frac{Q_c^4}{Q^4}$$

Fresnel reflectivity

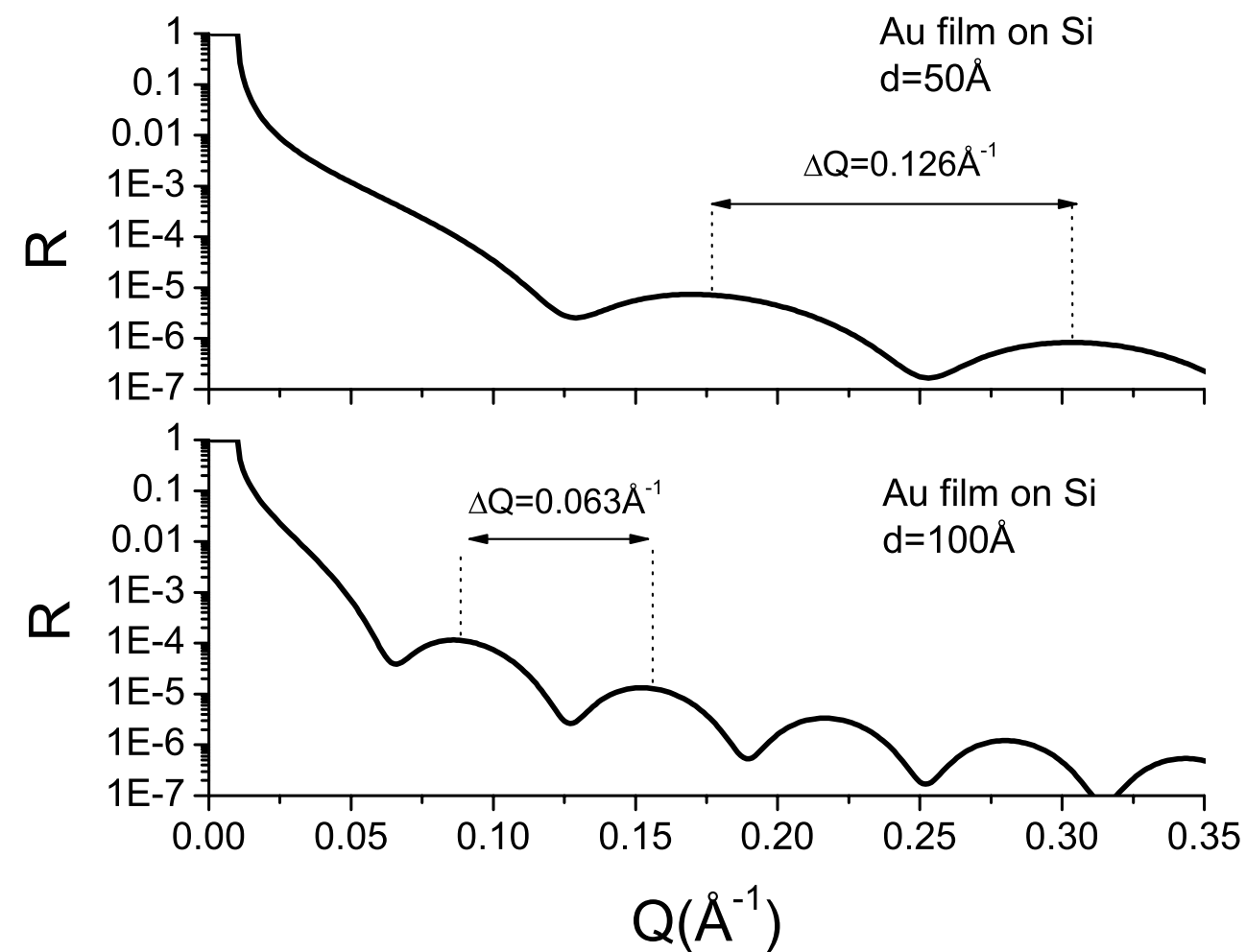
Summing up: neutron reflection at an interface

$$Q < Q_c \rightarrow R = 1$$

$$Q \gg Q_c \rightarrow R \approx \frac{1}{16} \frac{Q_c^4}{Q^4}$$

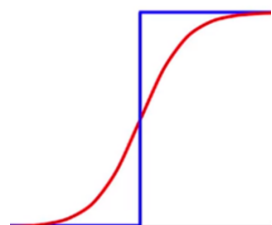


Single film reflectivity or why reflectivity is of practical interest



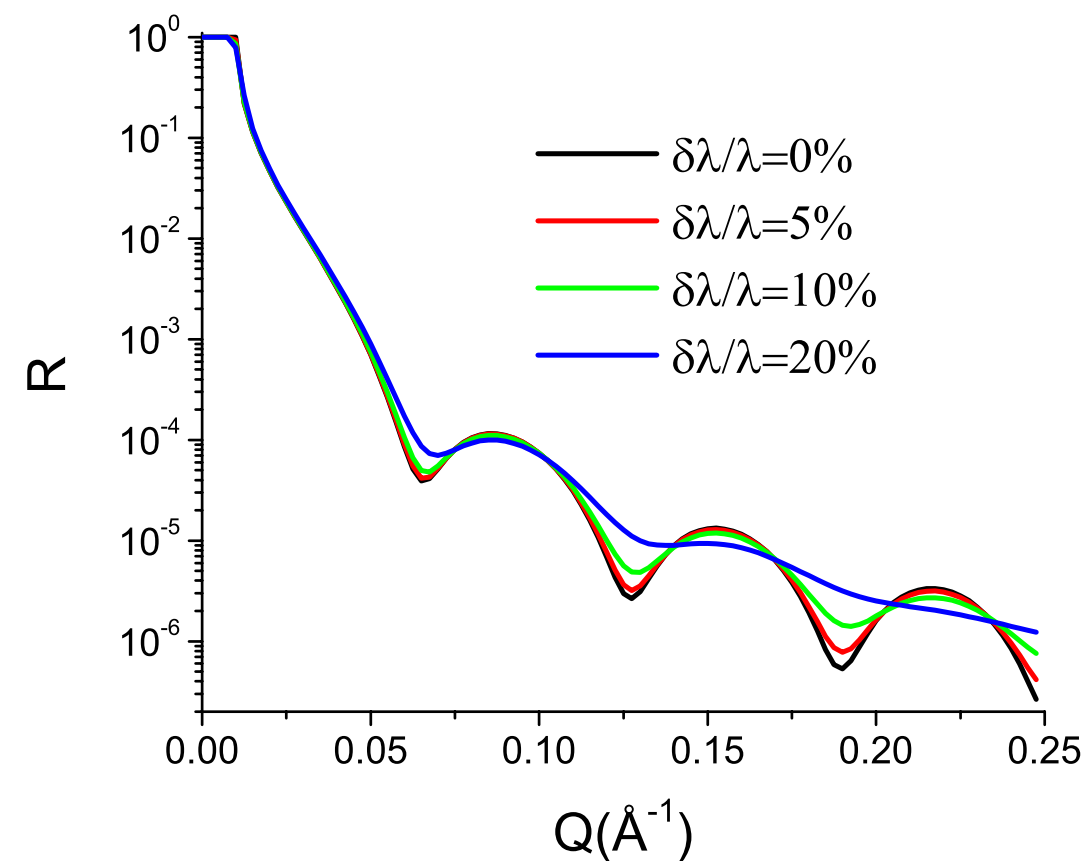
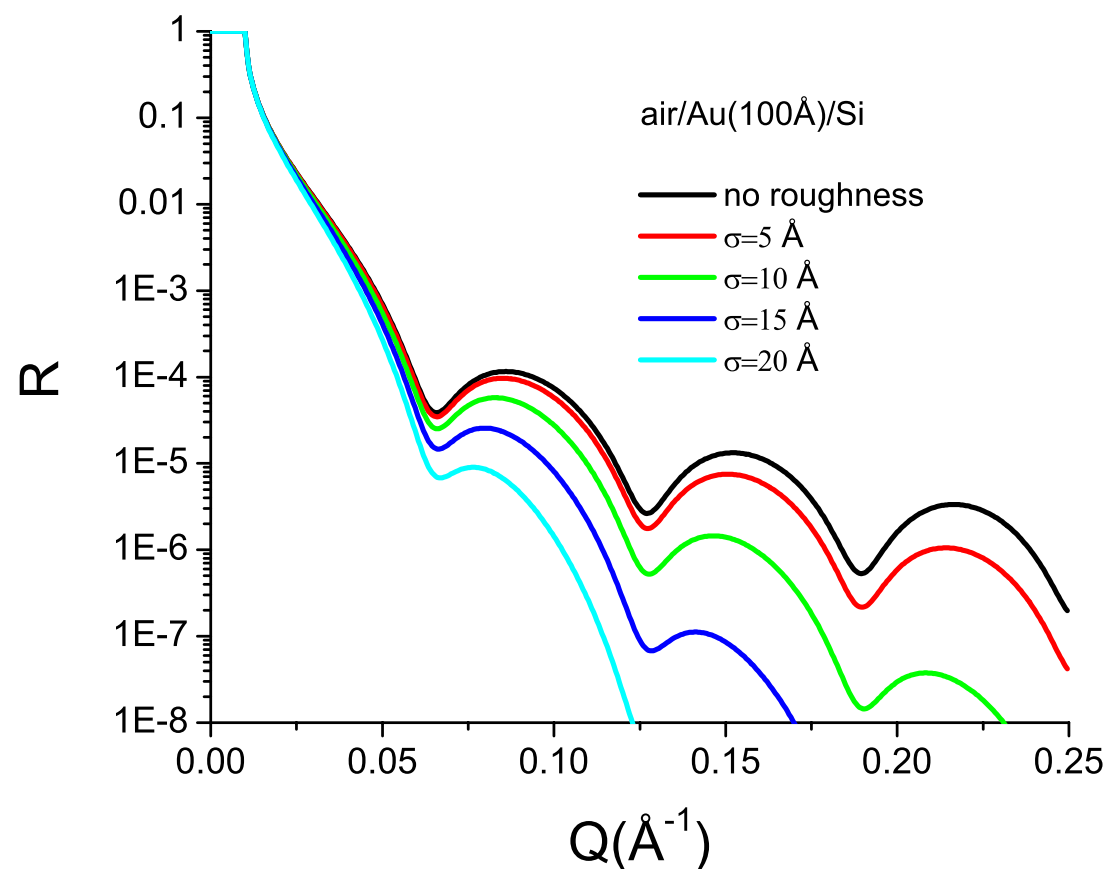
$$\Delta Q = \frac{2\pi}{d}$$

Two important additional details

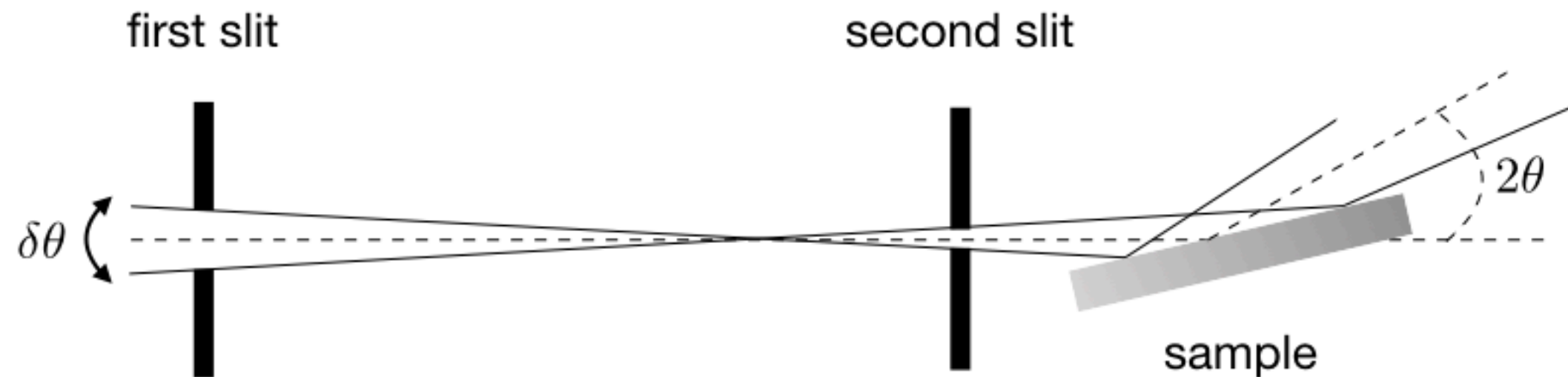


Layer roughness

Experimental Q-resolution



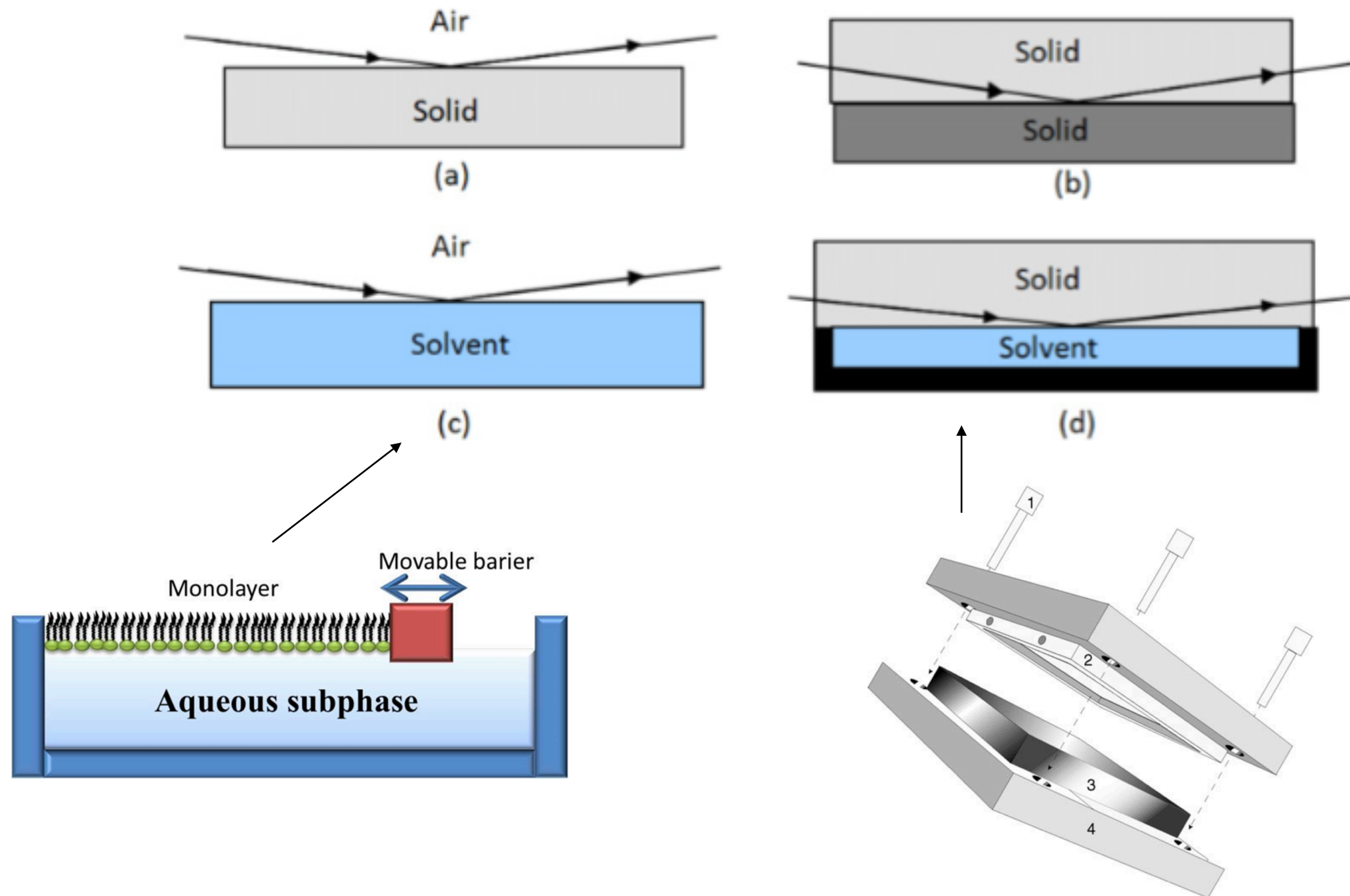
Experimental aspects



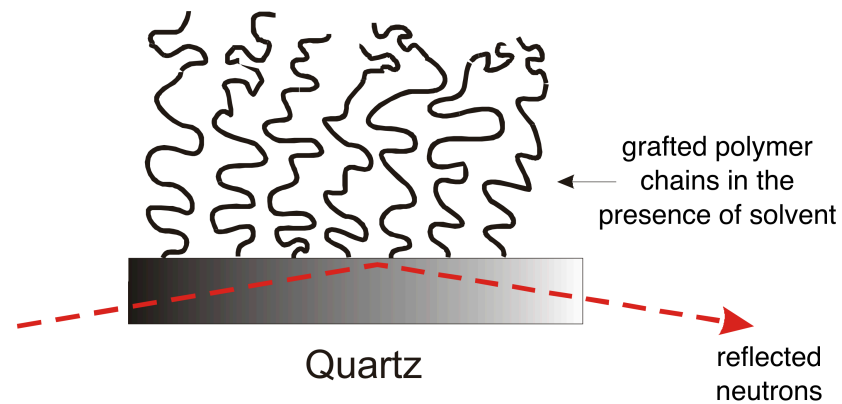
Two general types of reflectometers

- single wavelength (angle dispersive)
- time of flight (energy dispersive)

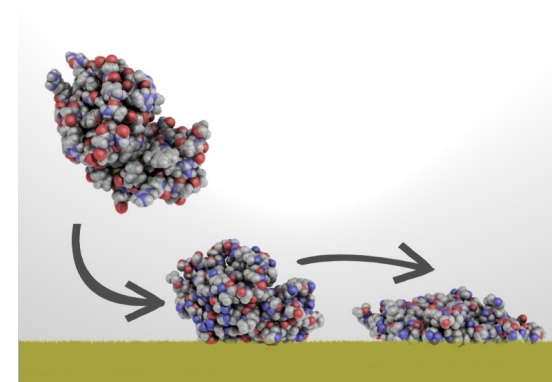
Experimental Setups and Sample environments



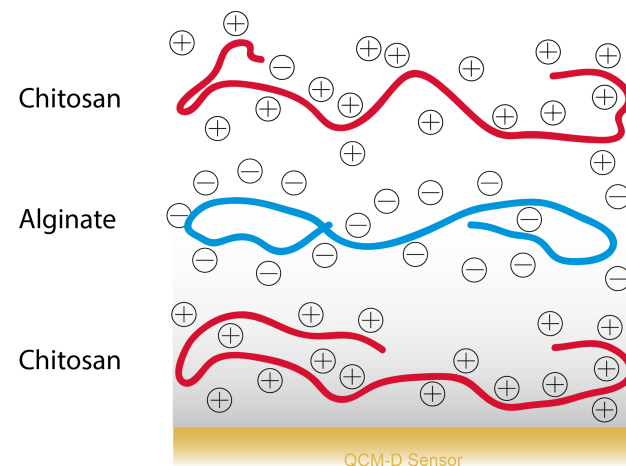
Typical systems that are studied by neutron reflectometry



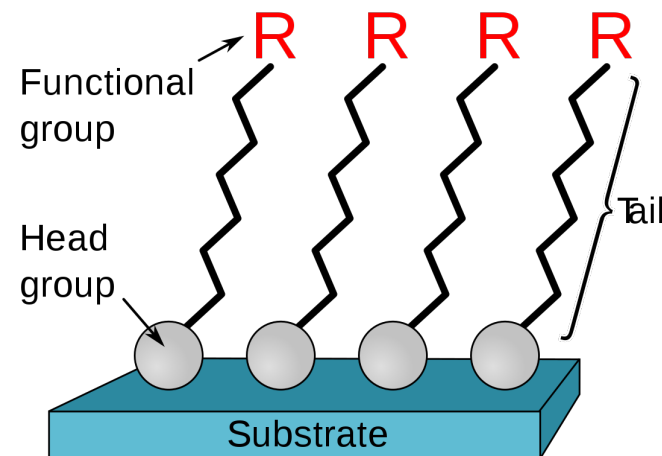
polymer brushes



adsorbed proteins

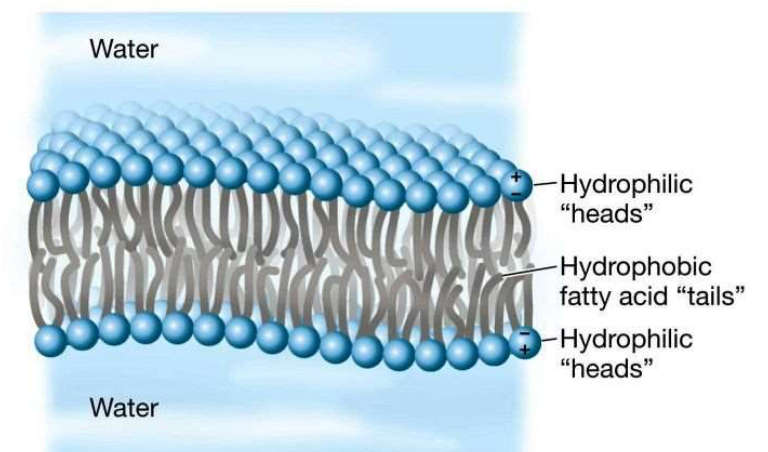


polyelectrolyte multilayers



self assembled monolayers

(B) Phospholipid bilayer

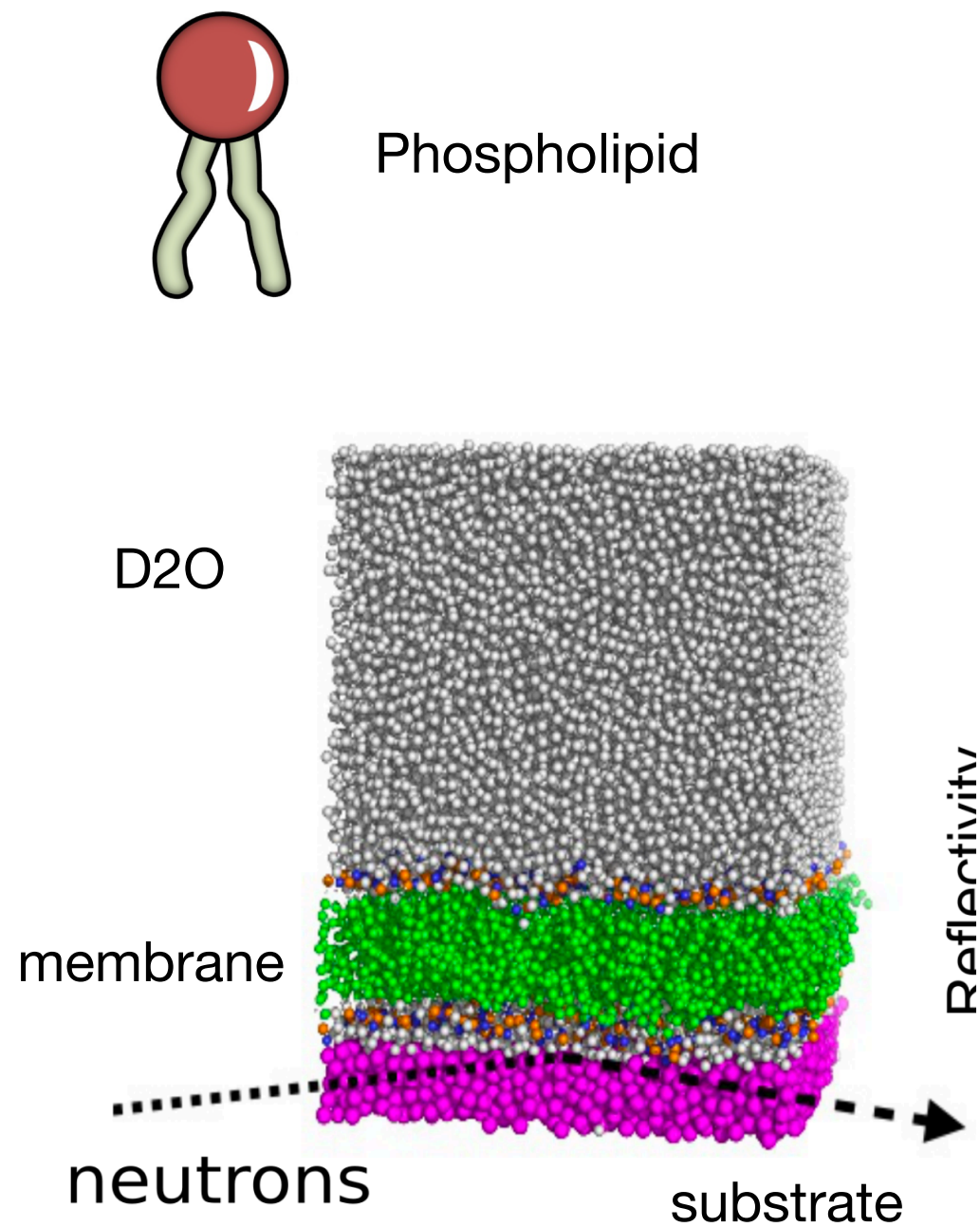


LIFE 8e, Figure 3.20 (Part 2)

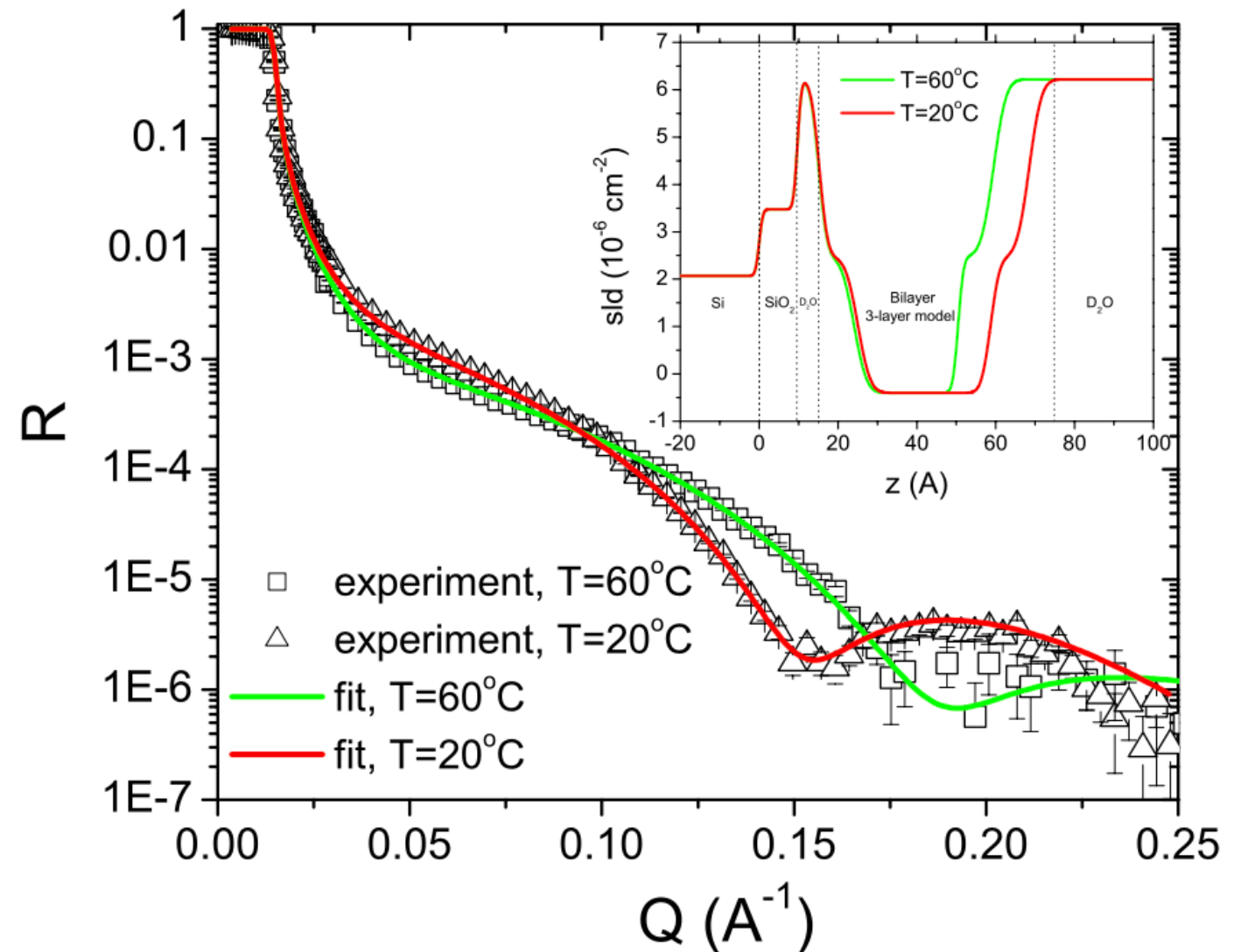
LIFE: THE SCIENCE OF BIOLOGY, Eighth Edition, © 2007 Sinauer Associates, Inc. and W. H. Freeman & Co.

biomembranes

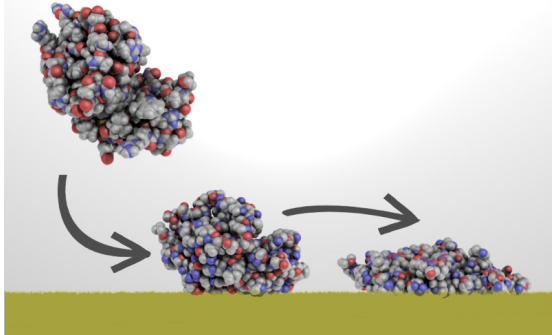
Lipid membranes at the solid/liquid interface



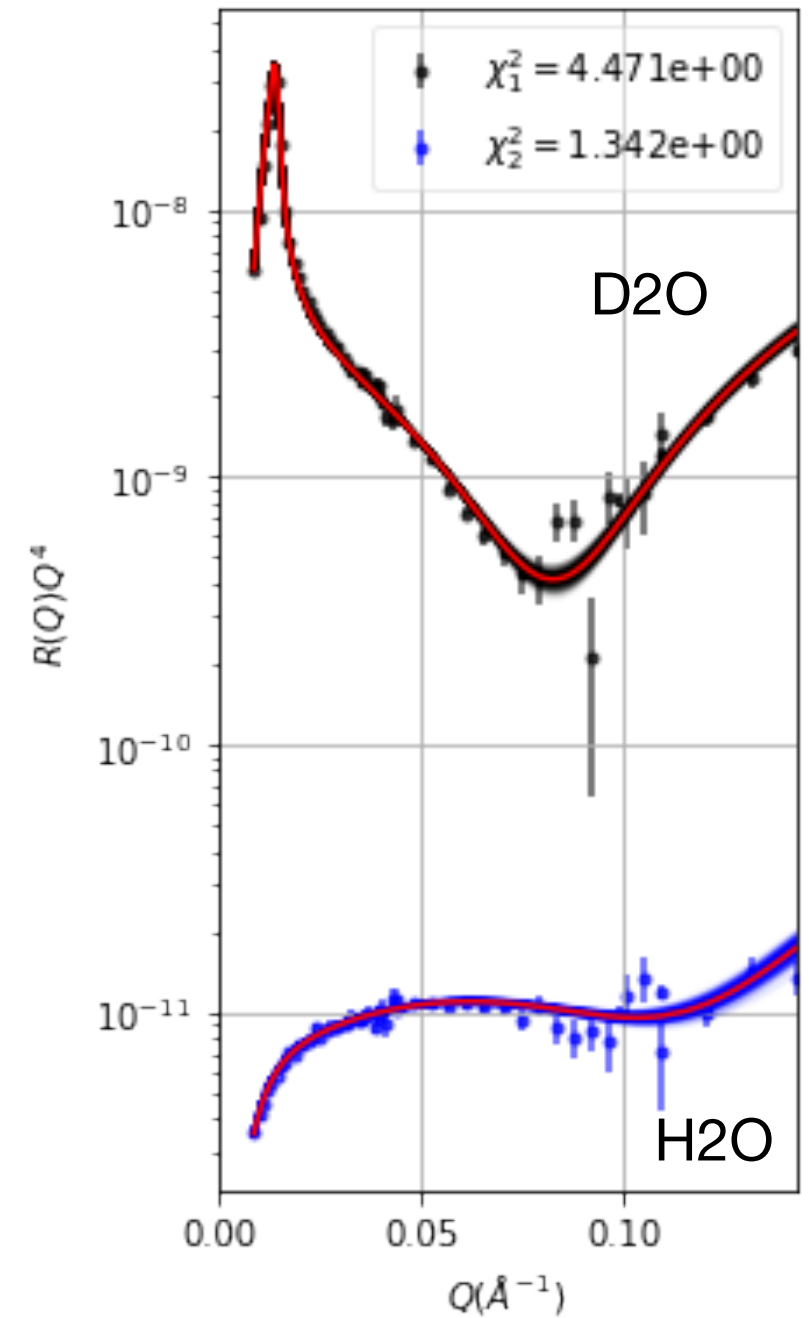
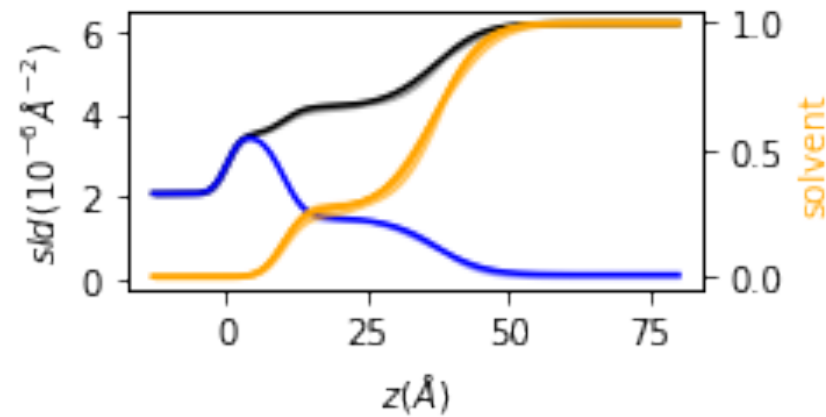
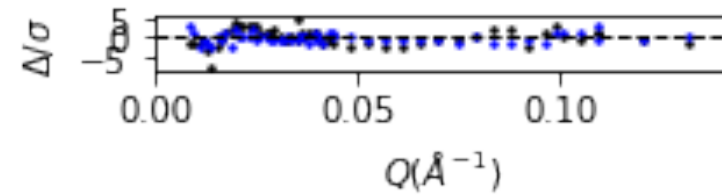
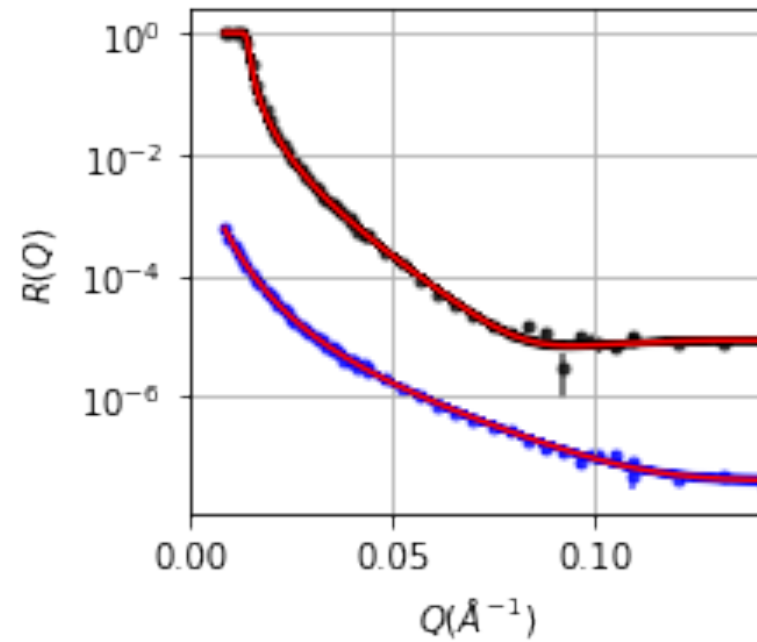
Temperature induced phase change



Proteins at the solid/liquid interface

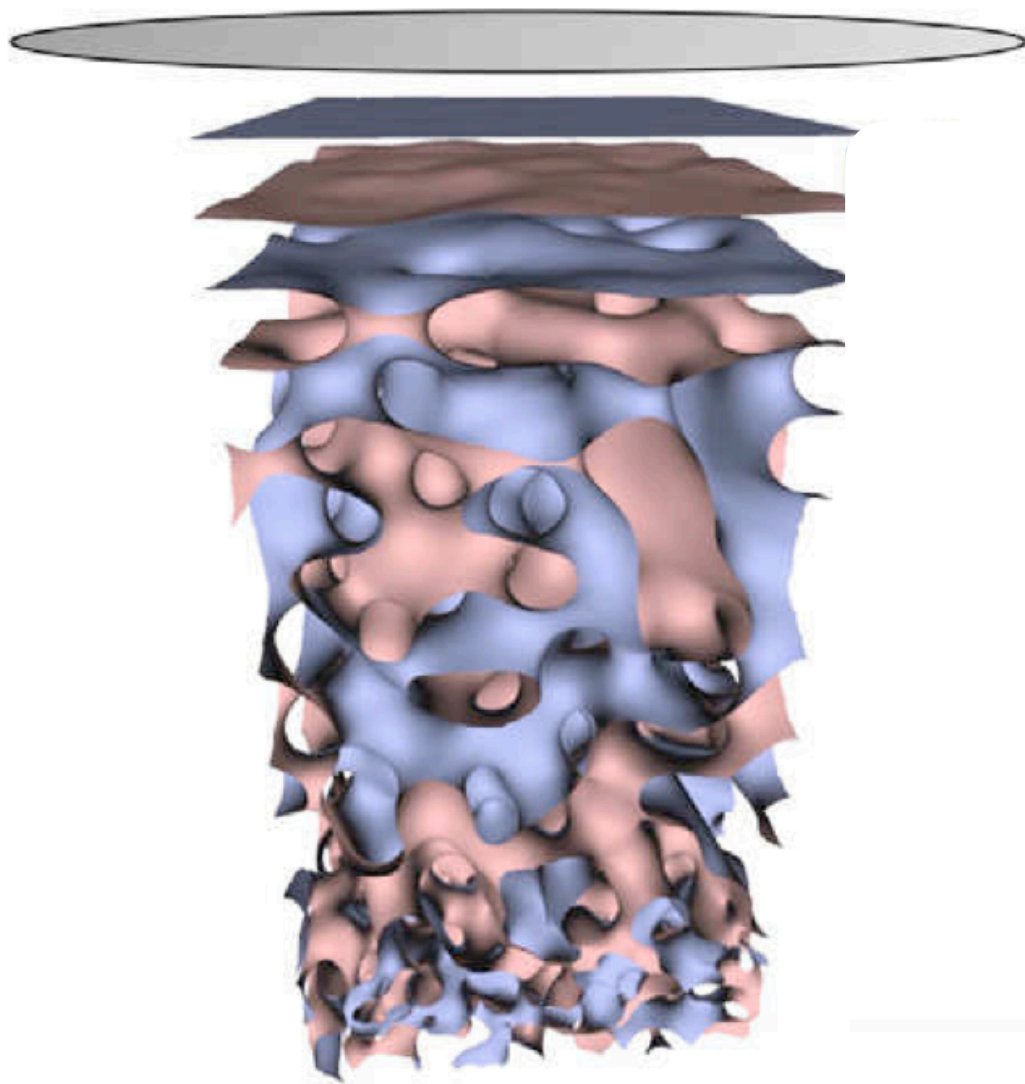


Lysozyme adsorption

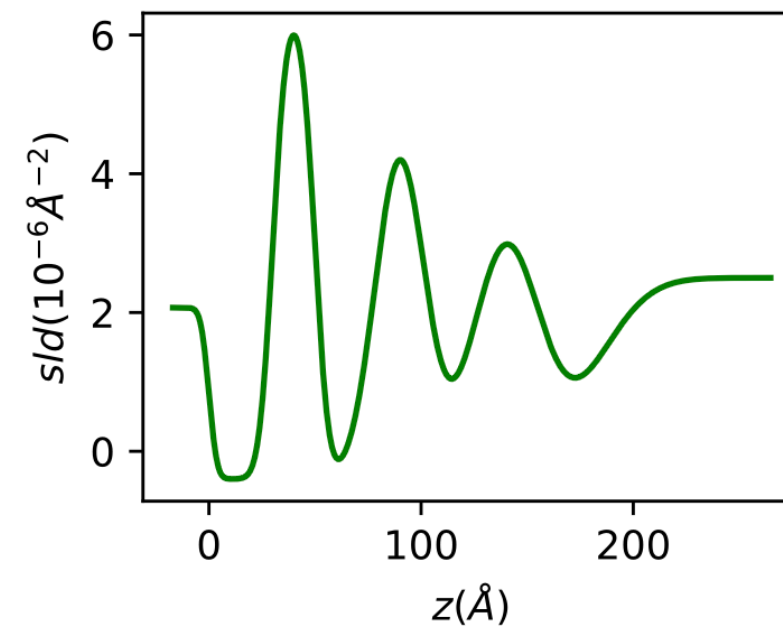
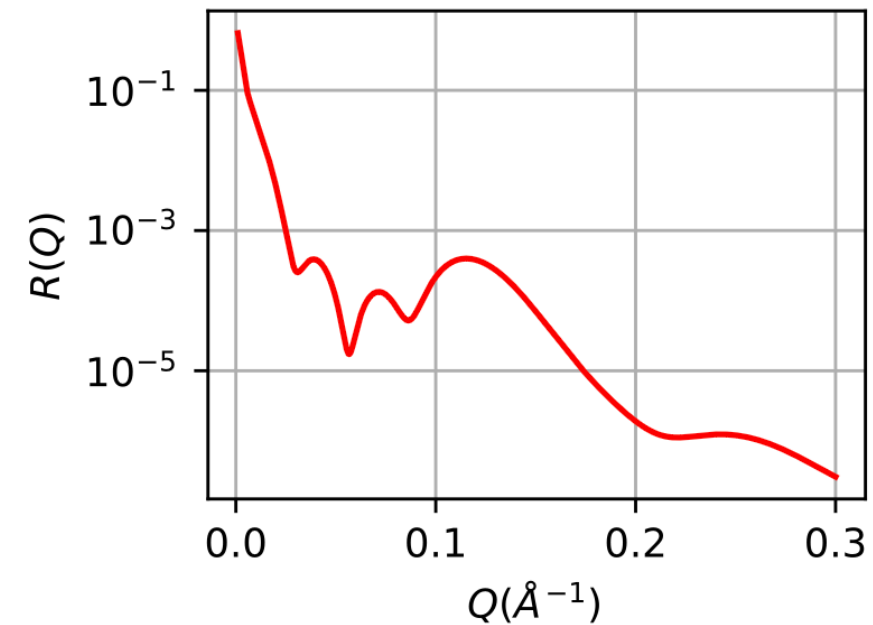


What it all has to do with food science?

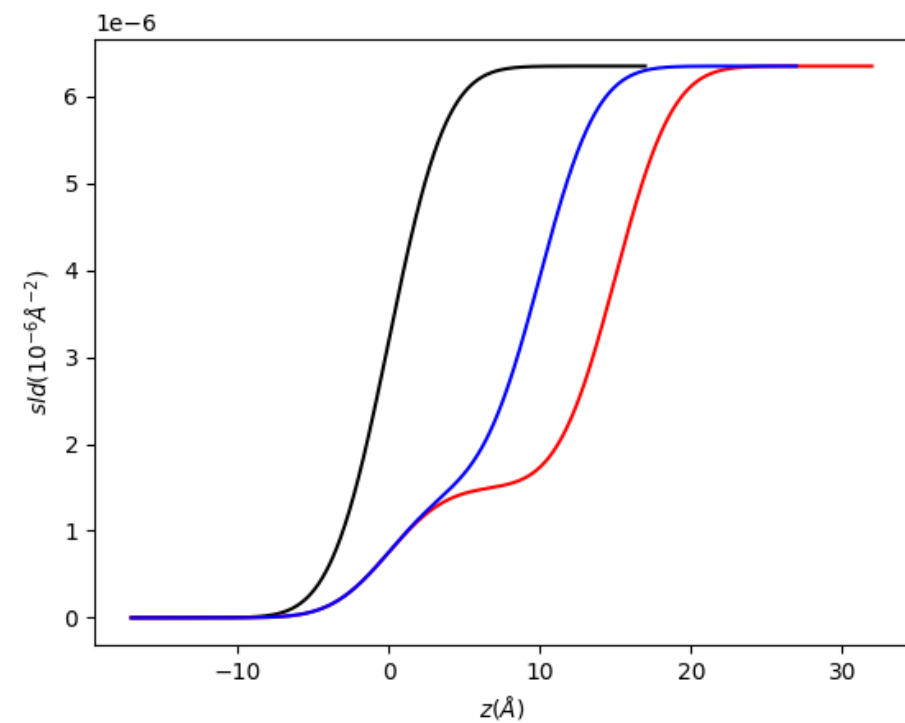
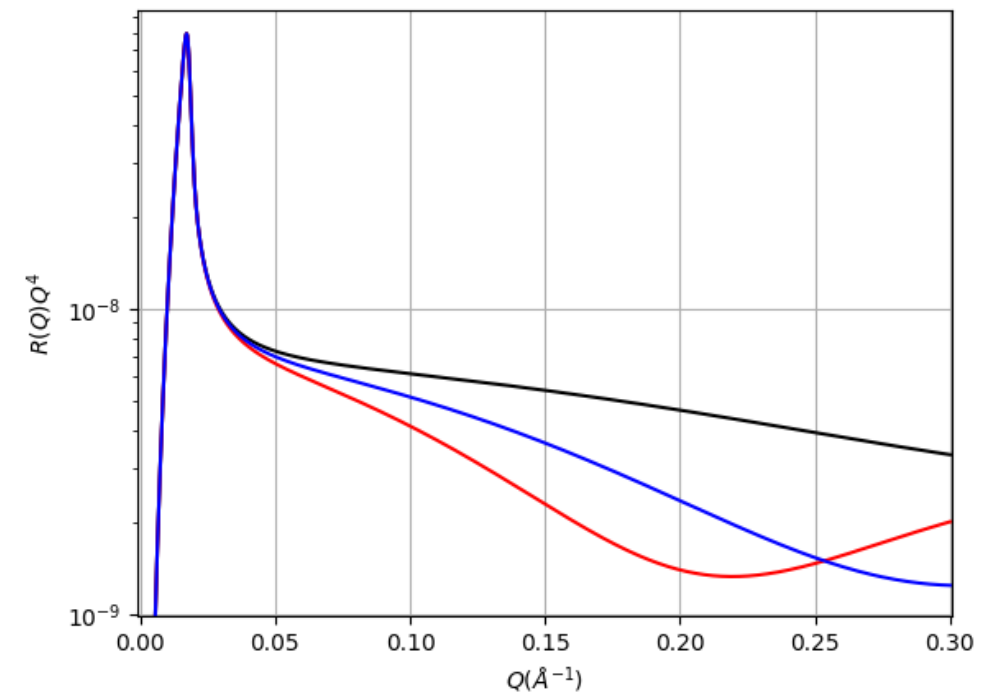
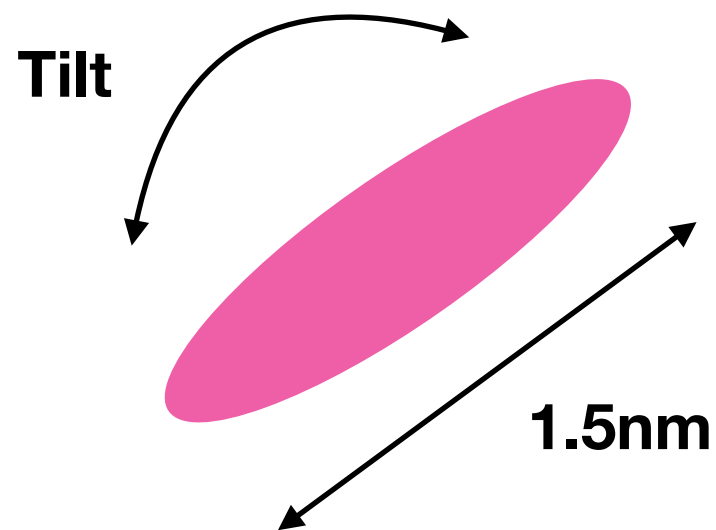
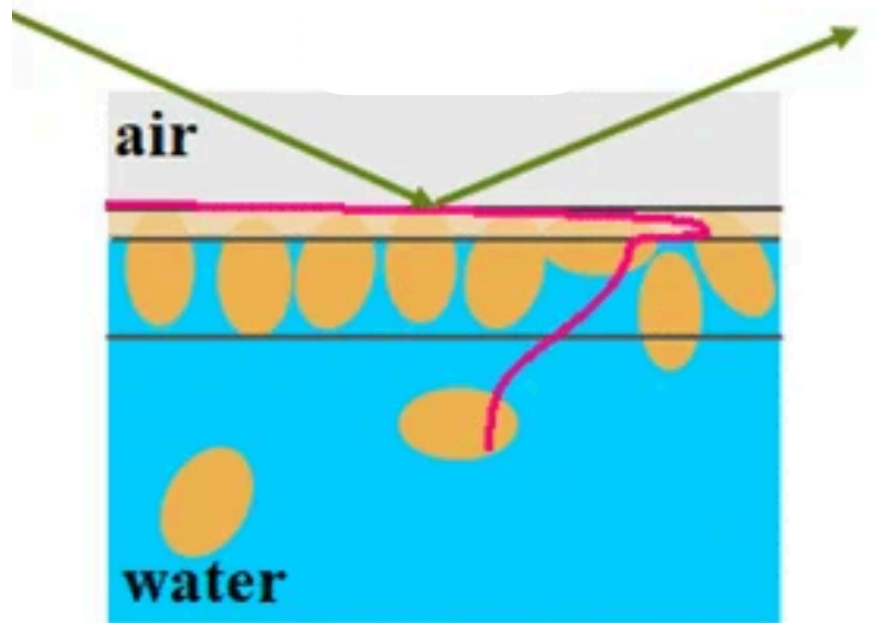
A first example. Bi-continuous microemulsions near surfaces



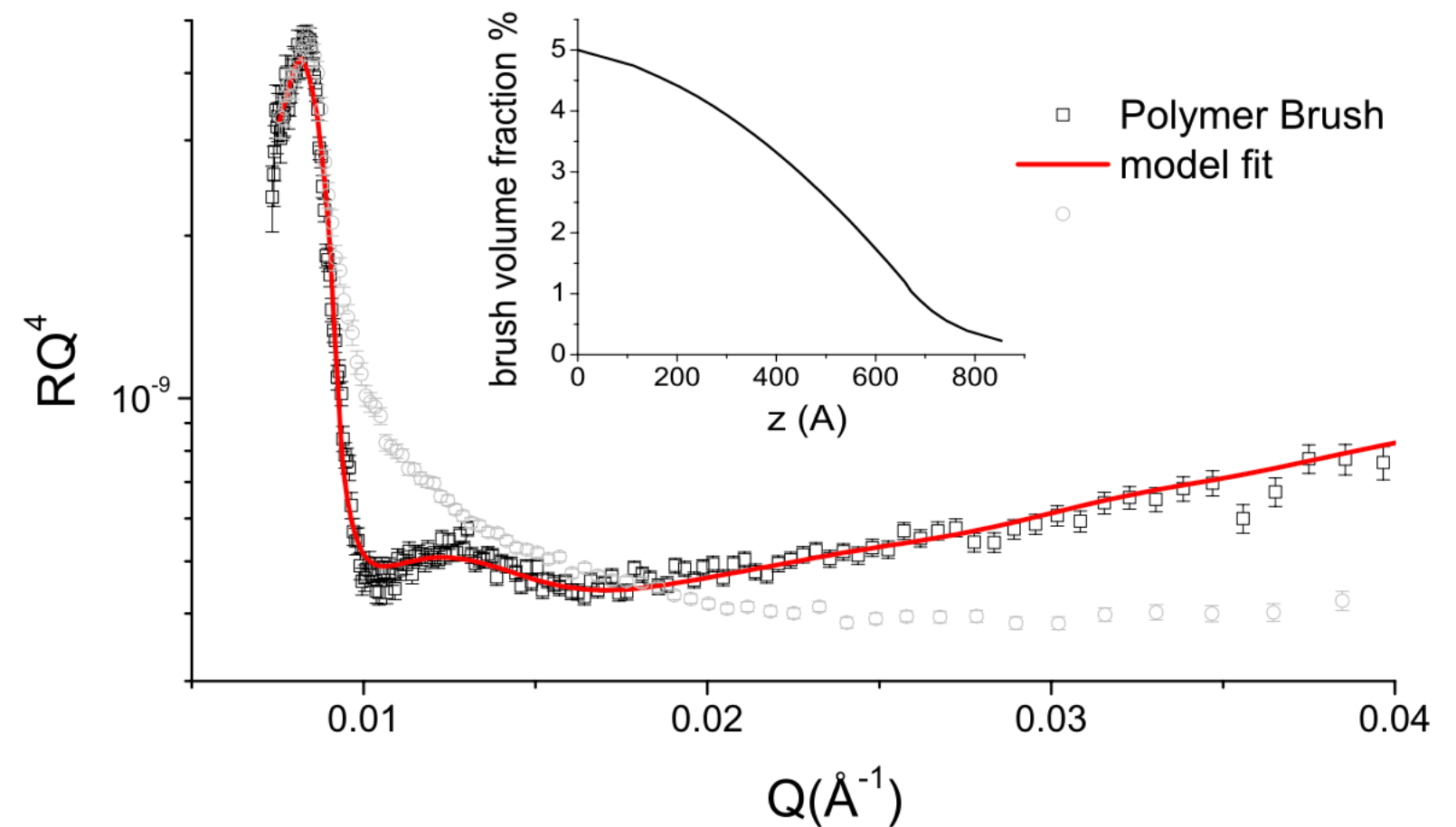
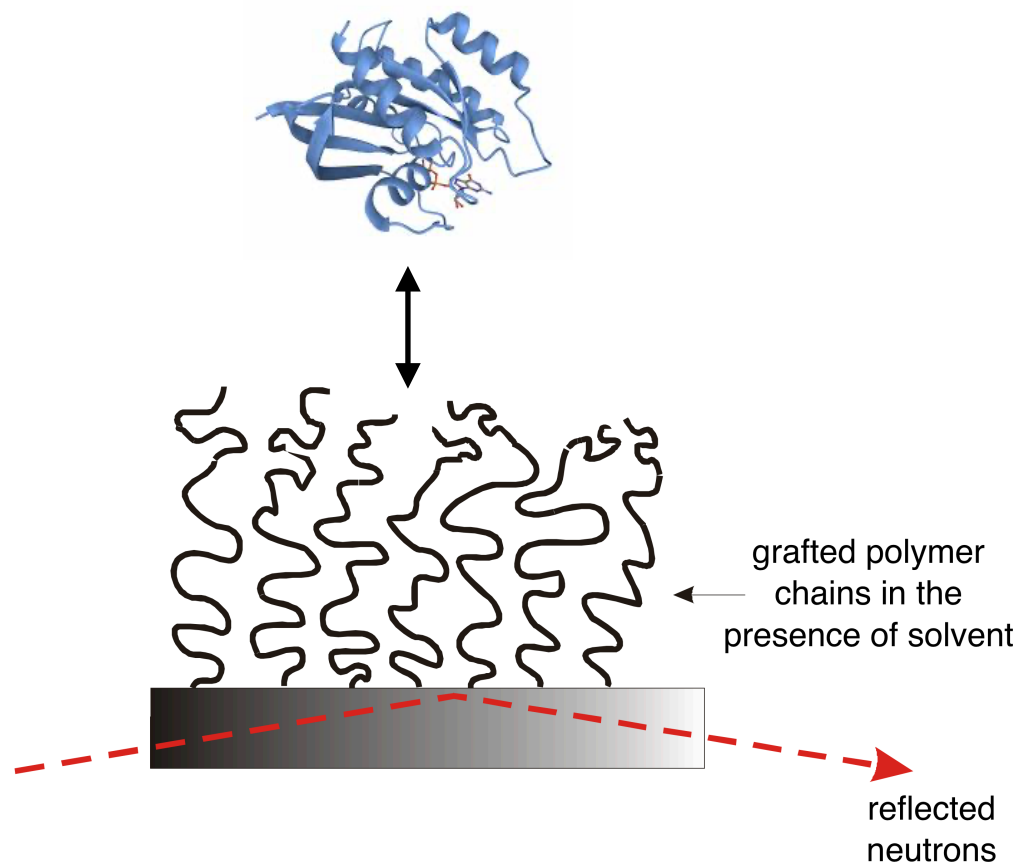
Lipfert et al. JCIS 534 (2019) 31



Another example. Proteins at the air or oil/water interface



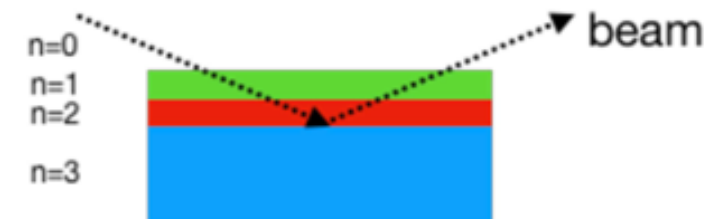
A final example. Protein repelling interface



Want to see more examples or play around with calculations on the cloud?

ANAKLASIS repository

<https://github.com/alexandros-koutsoumpas/anaklasis>



$$\left[\begin{array}{l} \left[\begin{array}{l} \text{Re_sld}_0, \text{Im_sld}_0, d_0, \sigma_0, \text{solv}_0, ds_0 \\ \text{Re_sld}_1, \text{Im_sld}_1, d_1, \sigma_1, \text{solv}_1, ds_1 \\ \text{Re_sld}_2, \text{Im_sld}_2, d_2, \sigma_2, \text{solv}_2, ds_2 \\ \text{Re_sld}_3, \text{Im_sld}_3, d_3, \sigma_3, \text{solv}_3, ds_3 \end{array} \right] , \\ \left[\begin{array}{l} \text{Re_sld}_1, \text{Im_sld}_1, d_1, \sigma_1, \text{solv}_1, ds_1 \\ \text{Re_sld}_2, \text{Im_sld}_2, d_2, \sigma_2, \text{solv}_2, ds_2 \\ \text{Re_sld}_3, \text{Im_sld}_3, d_3, \sigma_3, \text{solv}_3, ds_3 \end{array} \right] , \\ \left[\begin{array}{l} \text{Re_sld}_2, \text{Im_sld}_2, d_2, \sigma_2, \text{solv}_2, ds_2 \\ \text{Re_sld}_3, \text{Im_sld}_3, d_3, \sigma_3, \text{solv}_3, ds_3 \end{array} \right] , \\ \left[\begin{array}{l} \text{Re_sld}_3, \text{Im_sld}_3, d_3, \sigma_3, \text{solv}_3, ds_3 \end{array} \right] \end{array} \right]$$

Building models with Python lists...